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Unit – 3 : Session – 11 : SLO – 1 | Integrate Event Handling in GUI Applications

■ I. Fill in the Blanks

1. ActionListener — the interface used to handle button clicks in Swing.
2. addActionListener — `btn.addActionListener(listener)` registers the listener.
3. getActionCommand() — returns the action command string from an `ActionEvent`.
4. `java.awt.event` — the `ActionEvent` class lives in the `java.awt.event` package.

■ II. True or False

1. TRUE — Swing dispatches events synchronously on the Event Dispatch Thread (EDT), so long tasks should be moved off it.
2. TRUE — one `ActionListener` instance can be registered with any number of buttons (distinguished via `getActionCommand()`).

■ III. Multiple Choice

1. B — `actionPerformed(ActionEvent e)` is the single method of the `ActionListener` interface.
2. B — an `ActionEvent` object is passed to `actionPerformed()`.

■ IV. Programming

1. Output: C — clicking the button fires the lambda listener and prints "Hello".
2. Event-driven part: B — the anonymous `ActionListener` (its `actionPerformed` method) is the event-driven part of the snippet.